Every Kind of Analysis You Could Possibly Want to Do

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Data Exploration and Assessing Assumptions

1. Data Structure and Initial Inspection

2. Descriptive Statistics and Summary Measures

3. Distribution Analysis

4. Data Visualization and Exploratory Data Analysis

5. Missing Data Analysis

6. Outlier and Influential Observation Detection

7. Assessing Distributional Assumptions

8. Assessing Variance and Homogeneity Assumptions

9. Assessing Independence and Dependence

10. Assessing Relationships Between Variables

Comparing Groups (Means/Medians)

Relationships and Associations

Regression, Statistical Modeling and Prediction

Model Building, Diagnostics and Comparison

Visualization/ Presentation of Results

One-Sample t-Test

`Use when:`

Wilcoxon Signed-Rank Test

Predictive Modeling and Machine Learning

Time Series and Longitudinal Analysis

Survival and Event-Time Analysis

Multivariate and Dimension Reduction Techniques